# PROPERTIES OF THE CATEGORY OF QUASI-COHERENT MODULES ON THE SMALL ETALE SITE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Allowed standard prerequisites (editorial).*

- Compatibility of the equivalence with limits, colimits, kernels, cokernels, tensor products and internal homs. Same chapter, Lemma lemma-equivalence-quasi-coherent-limits, invoked at proof lines 3453-3454 and at each part. Carries each of the six properties across the equivalence between quasi-coherent modules on S and on S_etale.
- Preservation of finite presentability along the equivalence. Same chapter, Lemma lemma-equivalence-quasi-coherent-properties, invoked at proof lines 3505-3506. Used in part (6) to transfer the finite presentability hypothesis.
- The corresponding closure properties for quasi-coherent modules on a scheme. Schemes, Section schemes-section-quasi-coherent, invoked at proof lines 3451-3452. Supplies the facts on the scheme side that are being transferred.

*Source reference.* Original lemma 06VE: sources/stacks-descent/pinned-original-source.tex, source lines 3256–3256. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 05VG: sources/stacks-descent/pinned-original-source.tex, source lines 2830–2830. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original section 01LA: sources/stacks-chow/tags.txt, source lines 2079–2079. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $S$ be a scheme. The category $QCoh(S_{étale}, \mathcal{O})$ of quasi-coherent modules on $S_{étale}$ has the following properties:*

*Date*: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *Any direct sum of quasi-coherent sheaves is quasi-coherent.*
(2) *Any colimit of quasi-coherent sheaves is quasi-coherent.*
(3) *The kernel and cokernel of a morphism of quasi-coherent sheaves is quasi-coherent.*
(4) *Given a short exact sequence of $\mathcal{O}$-modules $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ if two out of three are quasi-coherent so is the third.*
(5) *Given two quasi-coherent $\mathcal{O}$-modules the tensor product is quasi-coherent.*
(6) *Given two quasi-coherent $\mathcal{O}$-modules $\mathcal{F}$, $\mathcal{G}$ such that $\mathcal{F}$ is of finite presentation. then the internal hom $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$ is quasi-coherent.*

*Proof.* The corresponding facts hold for quasi-coherent modules on the scheme $S$, see Schemes, Section 01LA. The proof will be to use Lemma 06VE to transfer these truths to $S_{étale}$.

Proof of (1). Let $\mathcal{F}_i$, $i \in I$ be a family of objects of $QCoh(S_{étale}, \mathcal{O})$. Write $\mathcal{F}_i = \mathcal{G}_i^a$ for some quasi-coherent modules $\mathcal{G}_i$ on $S$. Then $\bigoplus \mathcal{F}_i = (\bigoplus \mathcal{G}_i)^a$ by the lemma cited and we conclude.

Proof of (2). Let $\mathcal{I} \to QCoh(S_{étale}, \mathcal{O})$, $i \mapsto \mathcal{F}_i$ be a diagram. Write $\mathcal{F}_i = \mathcal{G}_i^a$ so we get a diagram $\mathcal{I} \to QCoh(\mathcal{O}_S)$. Then $\operatorname{colim} \mathcal{F}_i = (\operatorname{colim} \mathcal{G}_i)^a$ by the lemma cited and we conclude.

Proof of (3). Let $a : \mathcal{F} \to \mathcal{F}'$ be an arrow of $QCoh(S_{étale}, \mathcal{O})$. Write $a = b^a$ for some map $b : \mathcal{G} \to \mathcal{G}'$ of quasi-coherent modules on $S$. By the lemma cited we have $\operatorname{Ker}(a) = \operatorname{Ker}(b)^a$ and $\operatorname{Coker}(a) = \operatorname{Coker}(b)^a$ and we conclude.

Proof of (4). This follows from (3) except in the case when we know $\mathcal{F}_1$ and $\mathcal{F}_3$ are quasi-coherent. In this case write $\mathcal{F}_1 = \mathcal{G}_1^a$ and $\mathcal{F}_3 = \mathcal{G}_3^a$ with $\mathcal{G}_i$ quasi-coherent on $S$. By Lemma 06VE part (10) we conclude.

Proof of (5). Let $\mathcal{F}$ and $\mathcal{F}'$ be in $QCoh(S_{étale}, \mathcal{O})$. Write $\mathcal{F} = \mathcal{G}^a$ and $\mathcal{F}' = (\mathcal{G}')^a$ with $\mathcal{G}$ and $\mathcal{G}'$ quasi-coherent on $S$. By the lemma cited we have $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{F}' = (\mathcal{G} \otimes_{\mathcal{O}_S} \mathcal{G}')^a$ and we conclude.

Proof of (6). Let $\mathcal{F}$ and $\mathcal{G}$ be in $QCoh(S_{étale}, \mathcal{O})$ with $\mathcal{F}$ of finite presentation. Write $\mathcal{F} = \mathcal{H}^a$ and $\mathcal{G} = (\mathcal{I})^a$ with $\mathcal{H}$ and $\mathcal{I}$ quasi-coherent on $S$. By Lemma 05VG we see that $\mathcal{H}$ is of finite presentation. By Lemma 06VE part (8) we have $\mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) = (\mathcal{H}om_{\mathcal{O}_S}(\mathcal{H}, \mathcal{I}))^a$ and we conclude. □